Embryotoxic effects of pine needles and pine needle extracts.

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A heat stable toxin present in needles of ponderosa pine was found to be soluble in methanol, ethanol, chloroform hexanes and 1-butanol. The embryotoxic effects of fresh green pine needles and a chloroform/methanol extract were determined by measuring embryo resorption in pregnant mice. Autoclaving the needles and extract for 1 hour prior to feeding enhanced the embryoresorptive effect by 28% and 32%, respectively. The results of this study revealed that the embryo resorptive dose (ERD50) of heat stable toxin for 1 mouse was 8.95 gms. for fresh green pine needles and 6.46 gms. for autoclaved green pine needles. In addition to embryocidal effects, feeding of the toxin resulted in significant weight loss in adult mice.